TRACK A / TECHNICAL REPORT

# Lumen
# Coronary Evidence Explorer

Connect each coronary prediction to named anatomy and exact model reasoning. Challenge the demonstration with the complete development cohort, including false positives, missed disease and incorrect flags that persist across five input profiles.

**FINAL HOLDOUT EVALUATED**
Development and final test evidence are reported separately. All claims remain limited to the evaluated source dataset.

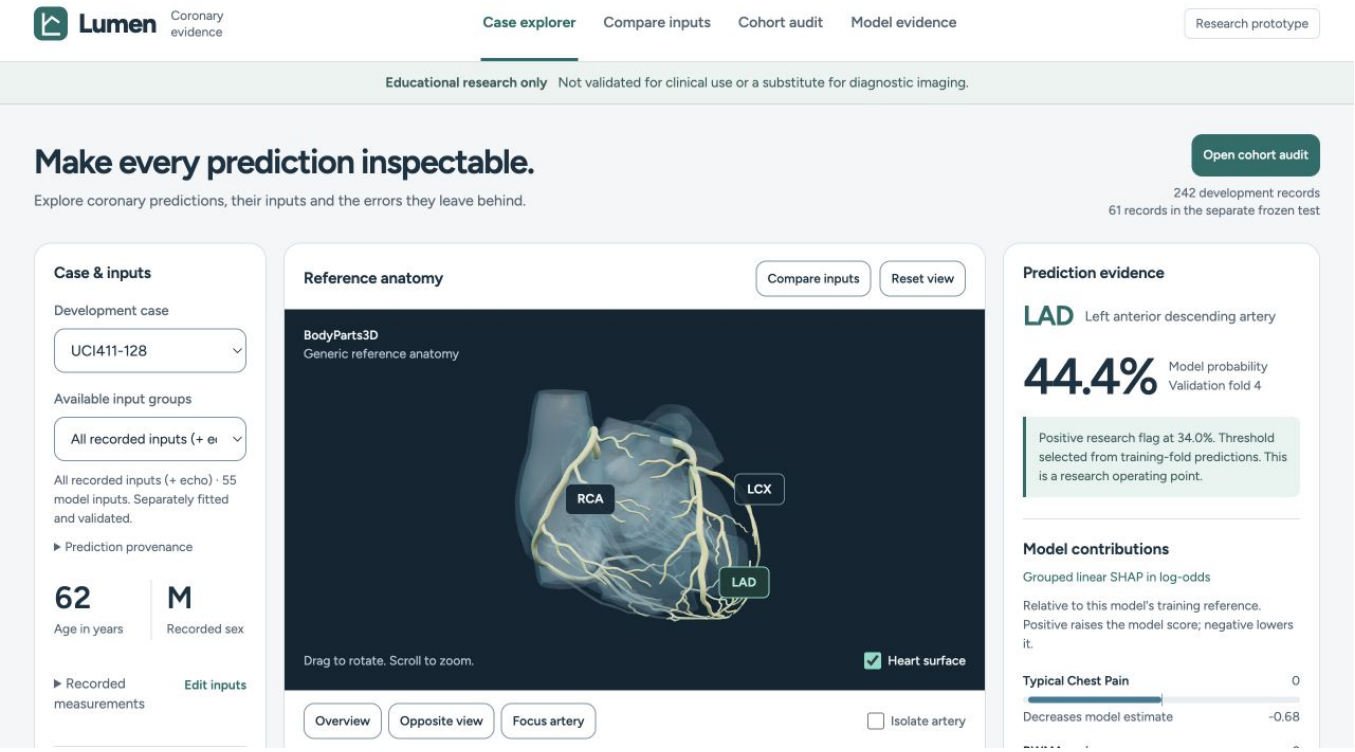



Figure 1. Actual running prototype. Generic BodyParts3D anatomy links named coronary targets to case inputs and explanations. The screenshot documents implemented interactions, not clinical validation.

| Clinical targets | Data partition | Deployment compute |
|---|---|---|
| CAD + LAD + LCX + RCA | 242 development<br>61 final holdout | CPU experiments + inference<br>Portable browser runtime |

**Contribution.** A reproducible audit connects all 242 development records to their excluding-fold models, five-profile comparisons, generic named arteries and exact explanations. Outcome-aware filters expose failures beyond the eight introductory examples. Source-bound receipts support inspection; this is an engineering contribution, not a claim of a new diagnostic algorithm.

Educational research prototype. Not validated for clinical use or a substitute for diagnostic imaging. Probabilities do not describe stenosis severity or patient-specific plaque locations.

01 / DATA AND VALIDATION

# Keep outcomes out of the inputs

**303 records**
Source hash checked

**242 development**
Nested model selection

**61 final holdout**
Scored once after freeze

All four target columns are excluded from every input pipeline.

**Source and targets.** The UCI extension of Z-Alizadeh Sani contains 303 records [1]. Cath supplies the overall CAD label; LAD, LCX and RCA supply vessel status. All four angiography columns are excluded from every task, leaving 55 predictors. Source hashes and frozen row identities are saved.

**Preprocessing.** Numeric median imputation and scaling, plus categorical imputation and one-hot encoding, are refitted within each training partition. Inference preserves the fitted values and reports unseen categories. Original source spelling is retained in the encoder.

| Input group | Count | Operational availability meaning |
|---|---|---|
| History | 17 | Demographics and recorded history indicators |
| Examination | 14 | Recorded symptoms and examination findings |
| ECG | 7 | Recorded ECG indicators |
| Laboratory | 14 | Laboratory measurements, source encodings retained |
| Echo | 3 | EF-TTE, Region RWMA and VHD |

The five availability groups are an explicit experiment definition informed by the source categories. Laboratory and echo are split operationally; the dataset does not establish a temporal care pathway. Verified units are used only where supported by source evidence [1,2].

**Nested selection (V2).** Five shared CAD-stratified outer folds evaluate the entire selection procedure. Three target-stratified inner folds select C (0.01, 0.1, 1, 10), unweighted/balanced logistic fitting, and optional sigmoid calibration by pooled inner log loss. Calibrated candidates use further cross-fitted training scores; outer validation records are excluded from fitting and calibration [4,5].

**Operating rule.** The selected candidate maximizes inner out-of-fold F1 over thresholds 0.05-0.95 at 0.01 steps. Ties favor proximity to 0.5, then the higher point. Outer classification metrics use the threshold fitted without each validation record. AUC and Brier use probabilities, independent of this classification threshold [6].

**Development audit:** 0 missing cells; 0 duplicated input rows; 1 CAD/vessel label discrepancy preserved. V2 followed inspection of V1 development results, so development evidence includes experiment-selection uncertainty.

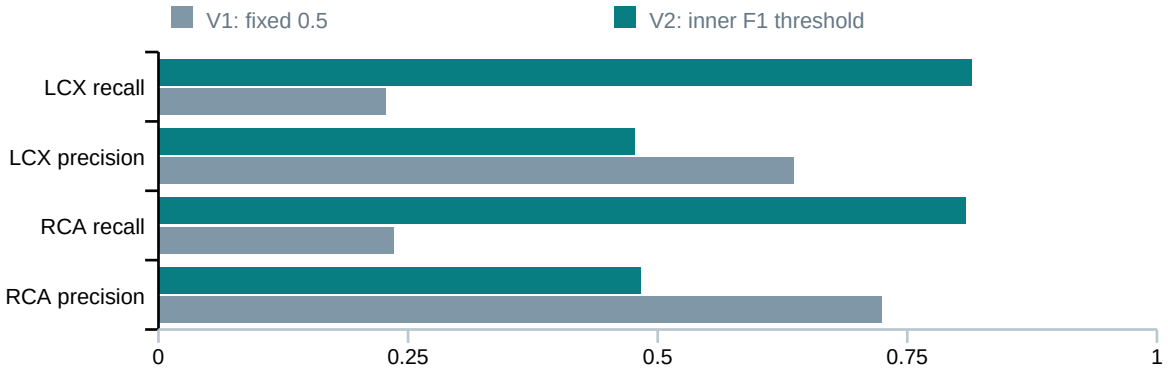
02 / PREDICTIVE EVIDENCE

# Detection improves with a cost

V2 pooled development out-of-fold results, n=242. Each record is predicted by an excluding outer model. Intervals bootstrap fixed OOF predictions (1,000 resamples); they omit refitting and experiment-selection uncertainty.

| Target | AUC | 95% AUC interval | Acc. | Prec. | Recall | F1 | Brier |
|---|---|---|---|---|---|---|---|
| CAD | 0.926 | 0.889 - 0.956 | 0.868 | 0.881 | 0.942 | 0.911 | 0.096 |
| LAD | 0.814 | 0.759 - 0.864 | 0.731 | 0.718 | 0.887 | 0.794 | 0.174 |
| LCX | 0.701 | 0.632 - 0.765 | 0.591 | 0.478 | 0.815 | 0.602 | 0.211 |
| RCA | 0.722 | 0.653 - 0.783 | 0.612 | 0.483 | 0.809 | 0.605 | 0.202 |

Figure 2. Lower fitted operating thresholds increase LCX/RCA recall while reducing precision. This is a classification tradeoff; vessel discrimination does not improve over the initial baseline.




| V2 selected rule | True positive | False negative | False positive | True negative |
|---|---|---|---|---|
| LCX | 75 | 17 | 82 | 68 |
| RCA | 72 | 17 | 77 | 76 |

The initial fixed-0.5 V1 baseline had LCX/RCA recall 0.228/0.236 and precision 0.636/0.724. V2 raises recall to 0.815/0.809 while precision falls to 0.478/0.483. The additional false positives remain visible, rather than being hidden behind a higher F1.

**Probability quality.** Brier and log loss are reported alongside ranking and classification metrics. A sigmoid step does not establish calibrated probabilities in a new population; Brier mixes reliability and discrimination, and the saved reliability bins are small [5].

These are development results, not external validation or medical operating thresholds. The four independent models do not impose a coherent joint CAD/vessel distribution.

03 / INPUT PROFILES AND COHORT AUDIT

# Agreement must survive a failure audit

Five cumulative input profiles are fitted separately under the same frozen development identities and outer folds. The full-input profile reuses V2. This evaluates model evidence with available groups; it does not estimate medical test utility, cost or causal benefit.

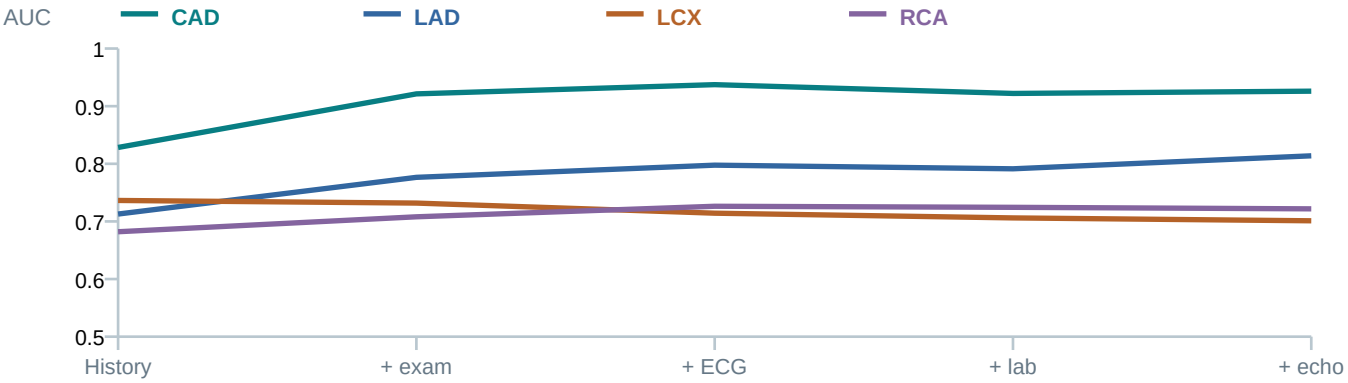



Figure 3. Development ROC-AUC by cumulative input profile. Each profile has its own nested fitting, calibration selection and threshold rule; groups are not merely nulled in the full-input model.

## All 242 records, including stable incorrect flags

| Target | Five profiles agree / 242 | Incorrect among agreeing | FP / FN |
|---|---|---|---|
| CAD | 182 | 12 | 10 / 2 |
| LAD | 161 | 33 | 32 / 1 |
| LCX | 172 | 60 | 51 / 9 |
| RCA | 164 | 58 | 56 / 2 |

Each flag uses its model's already fitted research threshold. Agreement selects a subset; it does not improve a retained prediction. LCX remains wrong on 60/172 agreeing records, and RCA on 58/164. These overlapping models are not independent witnesses.

The audit filters by target, outcome, profile agreement, outer fold and a shared fixed/research decision rule. Every selected record opens the anatomy and exact score. Recorded outcomes define filters: this is descriptive development review, not a blind or prospective test.

## Bounded model comparisons remain exploratory

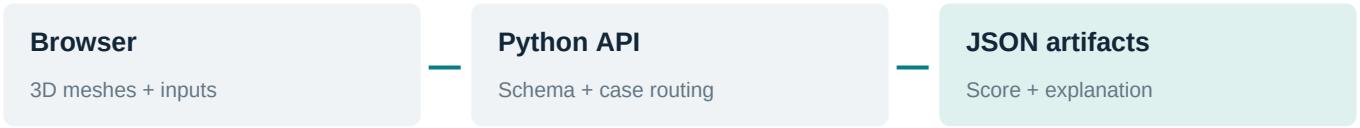
V1 and V3 retained prior/tree baselines. A later 200-fit nested CatBoost CPU probe took 190.33s and 423MB peak RAM: LAD AUC improved from 0.814 to 0.829, while CAD/RCA weakened and LCX was unchanged. These mixed development results did not replace the frozen V2 procedure or reopen final evaluation.

Measured elapsed time in the local CPU runs: V1 14.4s, V2 39.6s, V3 19.9s, availability 121.1s. V2 timing includes bounded search, export checks, conditional bootstraps and artifact creation. No GPU was used.

Model-family and input-profile decisions were frozen before final holdout scoring. Final outcomes were not used to retune the submitted procedure.

04 / ANATOMY AND INTEGRATION

# One mapping from score to vessel




Truth reveal is a separate API path. Edited inputs have no observed edited outcome.

**Reference geometry.** Official BodyParts3D memberships define heart FMA7088, LAD FMA3862, LCX FMA3895 and RCA FMA50039 [3]. Source OBJ parts are merged, triangulated and transformed into selectable Three.js groups [7]. Provenance retains hashes, memberships and modifications.

**Visual semantics.** The target selects matching geometry, probability and explanation. Overview, opposite view, mesh-bound focus and isolation complement rotate/zoom and the surface toggle. Isolation restores the chosen surface state. Probability colors the whole vessel; no patient geometry or lesion coordinates are inferred.

## Grouped interventional linear SHAP (log-odds)

For encoded values z, training means mu, coefficients beta and intercept b:

```
reference = b + sum(beta[j] * mu[j])
contribution[j] = beta[j] * (z[j] - mu[j])
score = reference + sum(contribution[j])
probability = sigmoid(score)
```

Encoded terms are grouped into raw inputs; calibration slope/offset enter coefficients/intercept. The waterfall shows reference, six largest terms, residual and final score. Sigmoid applies at the endpoint; sigmoid(reference) is not mean predicted probability.

Actual SHAP LinearExplainer with an Independent masker verifies the decomposition [9] using all encoded training-fold rows. Masks break feature dependencies and may form invalid one-hot states. Terms are interventional model log-odds, not conditional explanations or percentage-point probability changes.

Correlation-aware explanations can assign different credit. Terms show model associations; input edits explore fitted-model sensitivity, not treatment effects or clinical interventions.

## Verification and interaction

The compact export matched all 4,840 saved development probabilities exactly. It retains eight introductory cases plus the complete development audit. Contributions are calculated lazily from the same 100 profile/target/fold models; labels never enter inference.

The separate complete-cohort actual-SHAP certificate passed 4,840 explanations across 100 models; maximum grouped-term error 5.55e-17 and reconstructed score error 5.33e-15, tolerance 1e-12. Its Independent background excludes the record, the entire outer fold and every final holdout row. The original 160-explanation certificate remains scoped to the eight-case roster. These checks verify the specified decomposition, not clinical explanation utility.

Audit selections open the anatomy, score waterfall and five-profile matrix. Outcome-aware selection is disclosed; label reveal is separate from inference. Edits disable matching outcome comparison; unlabelled inputs use frozen full-development models.

**Current source verification, 12 October.** The 354-file redesigned working-tree export passed clean extraction: 20 Python tests, 52 Node subtests and 64 journey assertions. Current actual browser captures cover the excluded-fold LCX miss, complete score, five profiles, edited-case outcome guard and unlabelled inputs. These are AI software checks on one Mac, not human or clinical validation. Older fault/runtime checks remain dated historical evidence.

05 / FINAL EVIDENCE AND REPRODUCIBILITY

# Evidence boundaries and release record

Final holdout: n=61. The primary full-input procedure was frozen before scoring once. No holdout fitting, calibration, threshold or profile selection was performed.

| Target | AUC | 95% AUC interval | Acc. | Prec. | Recall | F1 | Brier |
|---|---|---|---|---|---|---|---|
| CAD | 0.908 | 0.815 - 0.972 | 0.836 | 0.851 | 0.930 | 0.889 | 0.110 |
| LAD | 0.893 | 0.804 - 0.966 | 0.852 | 0.814 | 0.972 | 0.886 | 0.126 |
| LCX | 0.748 | 0.621 - 0.859 | 0.607 | 0.535 | 0.852 | 0.657 | 0.216 |
| RCA | 0.724 | 0.579 - 0.844 | 0.590 | 0.500 | 0.760 | 0.603 | 0.209 |

Positive/negative counts: CAD 43/18; LAD 36/25; LCX 27/34; RCA 25/36. Fitted thresholds: CAD 0.21, LAD 0.36, LCX 0.30, RCA 0.25. Conditional intervals use 1,000 fixed-prediction patient resamples; no refitting uncertainty.

**LCX:** 23 TP, 4 FN, 20 FP, 14 TN. **RCA:** 19 TP, 6 FN, 19 FP, 17 TN.

## Limits

Small dataset; no external or prospective validation. A bounded cohort search admitted no full-55-input source: public AngioCAD lacks the examination and ECG groups [10]. LCX/RCA have limited discrimination and a large false-positive cost. Intervals omit refitting uncertainty. Independent targets can disagree. Generic anatomy cannot locate plaque; edits lack outcome labels.

## Reproduce and verify

Run `npm ci`, `python3 server.py`; open `http://127.0.0.1:8766/`. Python standard library or the static browser performs inference. Ordinary WebGL renders anatomy. `docs/cohort-audit.md` gives audit and pinned CPU verification commands.

**Local release identity.** Source ZIP: 354 payload files, SHA-256 f12fd1e42154f1d6...; static ZIP: 28 payload files, a5b667a63396402d.... Full hashes and the clean extraction receipt accompany the release. These are reviewed working-tree exports, not parity with the historical base commit. This PDF is a new presentation edition outside the frozen source ZIP. Public GitHub, YouTube and Devpost publication remain pending; second-device and successful Safari rendering checks are not established.

## Credits and AI assistance

UCI 411: Alizadehsani, R., Roshanzamir, M., & Sani, Z. (2013), CC BY 4.0 [1]. BodyParts3D, © The Database Center for Life Science, CC Attribution 4.0 International [3]. Geometry was merged, triangulated, translated and rotated. Three.js is MIT licensed [7].

OpenAI Codex assisted with research, code, experiments, debugging, AI review and documents. The redesign used UI/UX Pro Max and Anthropic Frontend Design guidance; Figtree is OFL-licensed. The human submitter is responsible for judging explanations; AI assistance must appear in Devpost Built With [8]. Human usability/signoff remain pending.

## Primary references

[1] UCI extension of Z-Alizadeh Sani. DOI: 10.24432/C5461K.
[2] Alizadehsani et al. Laboratory and echocardiography features (2013). DOI: 10.5812/cardiovascmed.10888.
[3] BodyParts3D archive: license and source download.
[4] scikit-learn: nested cross-validation.
[5] scikit-learn: probability calibration.
[6] scikit-learn: tuning the decision threshold.
[7] Three.js documentation.
[8] Multimodal AI Hackathon 2026: rules; supplied Track A and submission guidelines.
[9] SHAP LinearExplainer: interventional linear attribution.
[10] AngioCAD primary paper: 10.1016/j.cmpb.2026.109368; official feature workbook.